ELSEVIER

Physica B 197 (1994) 601–608

PHYSICA B

High-temperature ^{63,65}Cu NQR and NMR study of the high-temperature superconductor $La_{2-x}Sr_xCuO_4$ ($0 \leqslant x \leqslant 0.15$)

Takashi Imai[a,*], Charles P. Slichter[a], Kazuyoshi Yoshimura[b], Masaki Katoh[b], Koji Kosuge[b]

[a]*Department of Physics, University of Illinois at Urbana-Champaign, IL, USA*
[b]*Department of Chemistry, Kyoto University, Japan*

Abstract

The nature of strong correlation between carriers and its relation to the mechanism of superconductivity have been among the central issues in the study of high-temperature superconductors. Recently, we succeeded in observing ^{63,65}Cu NQR and NMR in the paramagnetic state of La_2CuO_4, an antiferromagnetic ($T_N \simeq 310$ K) parent material of the high-T_c superconductor $La_{1.85}Sr_{0.15}CuO_4$ ($T_c \simeq 38$ K). This enabled us to investigate the dependence of low-frequency Cu spin fluctuations on hole doping for the entire concentration range between the undoped insulating phase ($x = 0$) and the maximum T_c phase ($x = 0.15$) and to test scaling theories.

1. Introduction

A unique feature of high-temperature cuprate superconductors is that the electronic properties of the CuO_2 planes in the normal state above T_c are quite anomalous [1]. Understanding the anomalous normal state is believed to be a key to clarify the mechanism of high-T_c superconductivity. As noted first by Anderson [2], very strong correlation effects in the CuO_2 planes are considered to be the origin of the various anomalous experimental results which one can hardly understand in a standard Fermi liquid picture. As a matter of fact, the undoped parent materials such as La_2CuO_4 and $YBa_2Cu_3O_6$ are Mott insulators whose ground state is a Néel state, and a moderate carrier doping transforms these insulators to superconductors [1]. Obviously it is important to carry out comprehensive experiments covering the entire doping range between undoped insulators and moderately doped superconductors. In particular strong antiferromagnetic d-spin fluctuations which persist even in high-T_c superconducting phases [3,4] have to be fully understood to see if the spin fluctuation is the mechanism of Cooper pairing.

Nuclear quadrupole resonance (NQR) and nuclear magnetic resonance (NMR) of ^{63,65}Cu nuclear spins are very useful techniques, since one can observe the local electronic states at the planar Cu sites of high-T_c superconductors. These techniques have been applied to the study of a number of cuprates [5]. As a general tendency, however, strong paramagnetic antiferromagnetic spin fluctuations enhance the ^{63,65}Cu nuclear spin-lattice relaxation rate $1/T_1$ and the spin–spin relaxation rate $1/T_2$ below

* Corresponding author.

0921-4526/94/$07.00 © 1994 Elsevier Science B.V. All rights reserved
SSDI: 0921-4526(93)E0431-F

room temperature for lower doped materials. The enhancement of $1/T_2$ significantly limits the signal-to-noise ratio, and makes it difficult to observe nuclear resonance signals. Furthermore, the relatively high three-dimensional Néel temperatures, typically $T_N = 300\,K$–$500\,K$, require one to work at high temperatures, which also limits the signal intensity that is proportional to the Boltzmann factor. As a consequence, very few experiments had been reported for the paramagnetic state of lightly doped nonsuperconducting [6] or undoped [7] materials prior to the present work.

In this paper, we review our recent experimental attempt to study spin fluctuation properties of $La_{2-x}Sr_xCuO_4$ ($0 \leq x \leq 0.15$) by $^{63,65}Cu$ NQR/NMR [8,9]. We demonstrate the systematic change of the low-frequency part of the paramagnetic d-spin fluctuations between undoped insulating material La_2CuO_4 and a high-T_c superconductor $La_{1.85}Sr_{0.15}CuO_4$ ($T_c = 38\,K$). Following the brief reviews about the concepts of $1/T_1$ and $1/T_2$ in Section 2, and about the theoretical background of the two-dimensional (2D) quantum Heisenberg antiferromagnet in Section 3, we present clear evidence in Section 4 that one can understand the results of undoped La_2CuO_4 very well based on scaling theories for $S = 1/2$ 2D-Heisenberg model with nearest-neighbor exchange interaction $J \simeq 1500\,K$ as proposed by Chakravarty, Halperin and Nelson [10], and later by Chubukov, Sachdev and Ye [11] and Sokol and Pines [12]. In Section 5, we show our surprising finding that the hole doping into CuO_2 planes by the substitution of Sr^{2+} into La^{3+} does not change spin fluctuation properties significantly at elevated temperatures.

2. Concepts of $1/T_1$ and $1/T_2$

In this paper, we focus on two types of ^{63}Cu NQR/NMR properties. One is the nuclear spin-lattice relaxation rate (time) $1/T_1$ (T_1). T_1 is the time scale that describes how fast nuclear spin temperature relaxes from an excited state to the thermal equilibrium with the lattice by dissipating energy into it through electron–nucleus hyperfine interaction. Since the transition between Cu nuclear spin levels is caused by the fluctuating hyperfine magnetic fields from Cu electrons, one can indirectly sense Cu d-spin fluctuations by the measurements of T_1. Moriya [13] showed that $1/T_1$ is related to the transverse component of the imaginary part of the wave vector ($\boldsymbol{q}$) and frequency (ω) dependent dynamical electron spin susceptibility $\chi''_\perp(\boldsymbol{q}, \omega)$:

$$\frac{1}{T_1} = \frac{2k_B T}{g^2 \mu_B^2 \hbar^2} \sum_{\boldsymbol{q}} f_\perp(\boldsymbol{q})^2 \frac{\chi''_\perp(\boldsymbol{q}, \omega_n)}{\omega_n} \tag{1}$$

where $f_\perp(\boldsymbol{q})$ is the wave vector dependent form factor of hyperfine interaction given by $f_\perp(\boldsymbol{q}) = A_\perp + 2B[\cos(q_x a) + \cos(q_y a)]$, $A_\perp$ is the crystal a (b)-axis component of the hyperfine field at a nuclear spin generated by a Cu d-spin at the same site, B is the isotropic transferred hyperfine field generated by a nearest neighbor Cu d-spin [14], a is the lattice constant in the ab-plane, ω_n is the nuclear resonance frequency. Since $\omega_n \simeq 30\,MHz$ ($\simeq 1\,mK$) in our experiments, $1/T_1$ probes very low frequency spin fluctuations. It should be noted that the relevant physical quantity detected in inelastic neutron scattering experiments is the dynamical structure factor $S(\boldsymbol{q}, \omega) = \chi''(\boldsymbol{q}, \omega)/[1 + \exp(\hbar\omega/k_B T)]$.

Another quantity of our concern is the Gaussian component of the spin–spin relaxation rate $1/T_{2G}$. $1/T_{2G}$ is the square root of an appropriately defined second moment of nuclear spin–spin interaction. Equivalently, $1/T_{2G}$ is a measure of the homogeneous line width. Pennington et al. pointed out that RKKY-like indirect nuclear spin coupling enhances $1/T_{2G}$ an order of magnitude in high-T_c cuprates when the quantization axis of Cu nuclear spin is parallel to the crystal c-axis [15]. In this case, Pennington and Slichter [16] demonstrated that $1/T_{2G}$ is expressed by the $\boldsymbol{q}$-dependent static spin susceptibility $\chi'(\boldsymbol{q})$ as

$$\left(\frac{1}{T_{2G}}\right)^2 = \frac{0.69}{8\hbar^2}\left[\frac{1}{N}\sum_{\boldsymbol{q}} f_\parallel(\boldsymbol{q})^4 \chi'(\boldsymbol{q})^2 - \left(\frac{1}{N}\sum_{\boldsymbol{q}} f_\parallel(\boldsymbol{q})^2 \chi'(\boldsymbol{q})\right)^2\right] \tag{2}$$

where N is the number of Cu atoms in unit area. The form factor in Eq. (2) is given by

$$f_{\parallel}(\boldsymbol{q}) = A_{\parallel} + 2B[\cos(q_x a) + \cos(q_y a)] .$$

Since $A_{\parallel} \simeq -4B \simeq 160\,\mathrm{KOe}/\mu_B$ in La_2CuO_4 [8], $f_{\parallel}(\boldsymbol{q})$ is peaked at the corner of the first Brillouin zone $\boldsymbol{q} = \boldsymbol{Q} = (\pi/a, \pi/a)$. Therefore, to a very good approximation, $1/T_{2G}$ is proportional to the weighted $\boldsymbol{q}$-summation of $\chi'(\boldsymbol{q})$ around $\boldsymbol{q} = \boldsymbol{Q}$. When the energy scale of spin fluctuation is much smaller than temperature, the situation encountered in the case when critical slowing down occurs, the equal time correlation function $S(\boldsymbol{q})$ is related to $\chi'(\boldsymbol{q})$ as $S(\boldsymbol{q}) = \chi'(\boldsymbol{q})/k_B T$. The width of $S(\boldsymbol{q})$ around $\boldsymbol{q} = (\pi/a, \pi/a)$ is the inverse of the antiferromagnetic spin–spin correlation length ξ. Hence measurements of $1/T_{2G}$ provides information about ξ [16]. Combined information of $1/T_1$ and $1/T_{2G}$ provides knowledge about the relation between $S(\boldsymbol{q}, \omega)$ and $S(\boldsymbol{q})$ (i.e. $\chi''(\boldsymbol{q}, \omega)$ and $\chi'(\boldsymbol{q})$) and therefore about the characteristic energy scale of spin fluctuations.

3. Theoretical background of the 2D-Heisenberg system

A wide variety of experiments including neutron scattering experiments showed that the magnetic properties of La_2CuO_4 can be described very well based on 2D-Heisenberg model with a very weak interlayer coupling, which enhances the three-dimensional Néel temperature to ~300 K from $T_N = 0$ of the 2D system [1]. In the ground work of the 2D-Heisenberg antiferromagnet, Chakravarty, Halperin and Nelson proposed its phase diagram as shown in Fig. 1 [10]. They showed that there are two regimes in the paramagnetic state of the 2D-Heisenberg system when its ground state is a Néel state. One is the renormalized classical regime, $T \lesssim 2\rho_s$, and the other is the quantum critical regime, $T \gtrsim 2\rho_s$, where the zero-temperature spin stiffness constant ρ_s is given by $\rho_s = 0.18\,\mathrm{J} \simeq 300\,\mathrm{K}$ in La_2CuO_4. We summarize the expected NQR/NMR properties in these two regimes below.

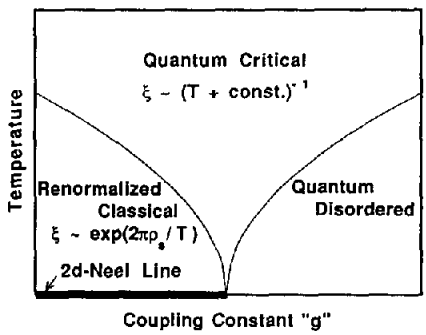


Fig. 1. Proposed phase diagram of a two-dimensional Heisenberg antiferromagnet (after Ref. [10]).

3.1. Renormalized classical regime ($T \lesssim 2\rho_s$)

In the renormalized classical regime, the correlation length ξ exhibits an exponential temperature dependence [10,17]

$$\frac{\xi}{a} = \frac{0.498}{1 + 0.5T/(2\pi\rho_s)} \exp\left(\frac{2\pi\rho_s}{T}\right) . \tag{3}$$

The dynamical scaling theory relates $S(\boldsymbol{q}, \omega)$ and $S(\boldsymbol{q})$ as

$$S(\boldsymbol{q}, \omega) = \frac{S(\boldsymbol{q})}{\omega_s} f(\boldsymbol{q}\xi, \omega/\omega_s) \tag{4}$$

where $S(\boldsymbol{q}) \simeq T^2\xi^2/(1 + \boldsymbol{q}^2\xi^2)$,

$$\omega_s = \frac{c}{\xi}\sqrt{\frac{T}{2\pi\rho_s}} \tag{5}$$

is the energy scale of spin fluctuations, and $f(\boldsymbol{q}\xi, \omega/\omega_s)$ is a dimensionless shape function [10]. When the d-spin system is cooled below $2\rho_s \simeq 600\,\mathrm{K}$, ω_s slows down exponentially (i.e. critical slowing down of the 2D system), then $S(\boldsymbol{q}, \omega)$ therefore $1/T_1$ diverges as calculated by Chakravarty and Orbach [18]. In the scaling limit $(T \to 0)$, it follows that

$$\frac{1}{T_1} \sim T^{3/2}\xi , \tag{6a}$$

$$\frac{1}{T_{2G}} \sim T\xi , \tag{6b}$$

and therefore

$$\frac{T_1 T}{T_{2G}} = T^{-1/2}. \tag{6c}$$

3.2. Quantum critical regime ($T \gtrsim 2\rho_s$)

In the quantum critical regime, the only energy scale is set by absolute temperature, and

$$\xi \sim \frac{1}{T + \text{const}} \tag{7}$$

holds instead of Eq. (3) [10,11]. Corresponding to Eq. (6), the following expressions hold [11,17]:

$$\frac{1}{T_1} \sim \text{constant}, \tag{8a}$$

$$\frac{1}{T_{2G}} \sim \xi, \tag{8b}$$

therefore

$$\frac{T_1 T}{T_{2G}} \sim \text{constant}. \tag{8c}$$

There is another regime called quantum disordered phase at low temperatures in Fig. 1. Because of frustration effects caused by exchange interaction with second-nearest-neighbor, by the presence of doped holes, etc., one may have a spin-glass-like ground state instead of a Néel state. In this phase, the correlation length ξ saturates at low temperatures, and the spin excitation spectrum has a gap.

4. Undoped La_2CuO_4

Presented in Fig. 2 are the temperature dependence of $1/T_1$ and $1/T_{2G}$ for La_2CuO_4 measured by NQR for a powder ceramic sample sealed in vacuum [9]. We found for an unsealed powder sample that a tiny amount of absorption of oxygen of order of 0.0035 atom per unit cell in the warming process in air results in the extrinsic enhancement of $1/T_1$ due to quadrupolar processes below 700 K after a heat cycling [8]. We also observed motional narrowing effects on $1/T_2$ above 700 K. Therefore it is essential to seal

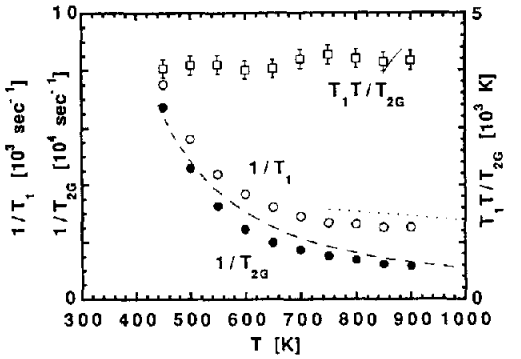


Fig. 2. Temperature dependence of $1/T_1$ and $1/T_{2G}$ observed for La_2CuO_4. Dotted and dashed curves indicate the theoretical prediction for $1/T_1$ and $1/T_{2G}$ based on the 2D-Heisenberg model after Refs. [21] and [22], respectively. Note that there is no adjustable parameter in these theoretical calculations.

the sample in vacuum to assure reproducibility. For further details of experimental procedures, we refer readers to our original papers [8,9].

Evidently, both $1/T_1$ and $1/T_{2G}$ diverge with lowering temperature as expected for the 2D-Heisenberg system. In Fig. 3, we present $1/T_1 T^{3/2}$ and $1/T_{2G}T$ as a function of inverse temperature in a logarithm scale. As expected from Eq. (3) and Eq. (6a,b), we observe nearly linear behavior at low temperatures, where the slope of the plotted data is determined by J. In fact we obtain $J = 0.138 \pm 0.12$ eV [8] and $J = 0.132 \pm 0.05$ eV [9] from the fit of $1/T_1$ and $1/T_{2G}$, respectively, to the rigorous expressions

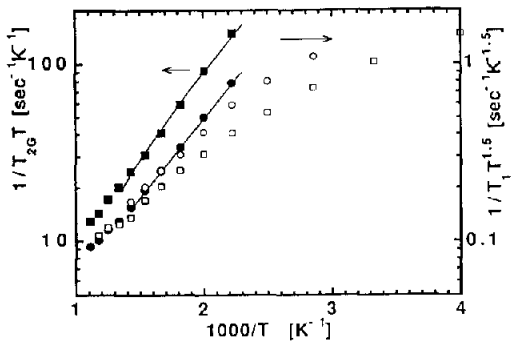


Fig. 3. Temperature dependence of $1/T_1 T^{1.5}$ for $x = 0$ (●), $x = 0.02$ (○) and $x = 0.04$ (□), and $1/T_{2G}T$ for $x = 0$ (■). Solid curves are the best fit to the rigorous expression of $1/T_1$ given in Ref. [18] with $J = 0.135$ eV and $1/T_{2G}$ given in Ref. [9] with $J = 0.132$ eV.

valid in the renormalized classical regime. The estimated value of J agrees excellently with the value determined by high-energy neutron scattering experiments by Hayden et al., namely $J = 0.132$ eV [19].

At higher temperature, $1/T_1$ levels off to $\sim 2600\,s^{-1}$, in agreement with the estimated value $\sim 3300\,s^{-1}$ of Eq. (5) by Chubukov and Sachdev based on the $1/N$ expansion technique applied to the quantum critical regime [11]. We note that theoretical calculations based on Gaussian approximations by Gelfand and Singh [20] and finite-cluster calculations by Sokol, Gagliano and Bacci [21] resulted in nearly temperature independent $1/T_1$ without relying on the scaling argument. Calculations of $1/T_{2G}$ using a high-temperature expansion technique by Sokol, Glenister and Singh [22] also showed excellent agreement with our data.

In Fig. 4, we show the temperature dependence of ξ estimated from our results of $1/T_{2G}$ [9]. The crossover from exponential temperature dependence at low temperatures as reported by Keimer et al. [23], Eq. (3), to linear temperature dependence, Eq. (7), is robust.

From these results, we conclude that the theoretical description of the ^{63}Cu NQR data of La_2CuO_4 based on the 2D-Heisenberg model works very well both in the renormalized classical and quantum critical regimes. However, the fact that the ratio T_1T/T_{2G} shown in Fig. 2 is temperature independent in the entire temperature range disagrees with Eq. (6c), and suggests that the crossover from the renormalized classical regime to the quantum critical regime occurs gradually in a broad temperature interval [9].

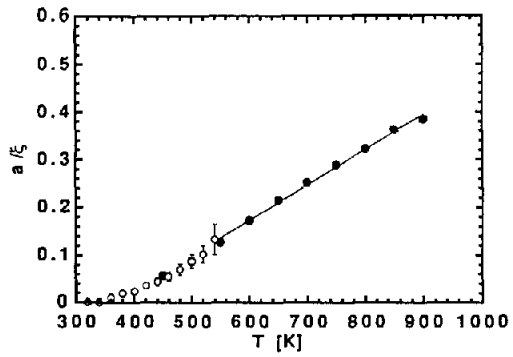


Fig. 4. Temperature dependence of the inverse correlation length a/ξ in La_2CuO_4 estimated from the results of $1/T_{2G}$ presented in Fig. 2 (●), compared with the results of neutron scattering experiments by Keimer et al. (○).

5. Influence of hole doping

In Fig. 5(a), the temperature dependence of $1/T_1$ is presented for various Sr concentrations [8]; $x = 0.02$ (doped insulating antiferromagnet, $T_N = 73$ K), 0.04 ('spin-glass' phase), 0.075 (superconductor, $T_c = 23$ K) and 0.15 (superconductor with highest T_c, $T_c = 38$ K). Also shown in Fig. 5(b) is the temperature dependence of T_1T. The physical meaning of T_1T may be understood as follows. If we use the scaling argument following Sokol and Pines [12], one may write $\chi(q, \omega) \sim \xi^2 \chi(q\xi, \omega/\omega_s)$. Then, integrating Eq. (1), it follows that $T_1T \sim \omega_s$, the

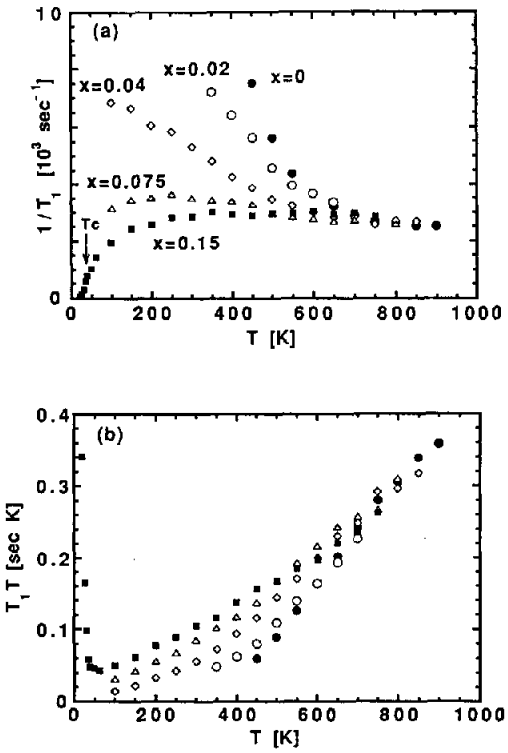


Fig. 5. Temperature dependence of (a) $1/T_1$ and (b) T_1T for various doping levels of $La_{2-x}Sr_xCuO_4$. The symbols in (b) are the same as in (a).

characteristic frequency of spin fluctuations. If we further assume that the dynamical critical exponent $z = 1$, i.e. $\omega_s \sim \xi^{-z} = \xi^{-1}$, which is confirmed to be valid for La_2CuO_4 in Section 4, then $T_1T \sim \xi^{-1}$.

The most striking feature in Fig. 5(a) is that $1/T_1$ merges to essentially the same value of $\sim 2750 \pm 150\,s^{-1}$ regardless of the amount of hole doping. It should be recalled that nearly temperature independent $1/T_1 \sim 2600\,s^{-1}$ observed around 700 K for La_2CuO_4 is accounted for both by quantum critical scaling [11] and numerical calculations for the 2D-Heisenberg model [20,21]. This immediately leads us to the very important and surprising conclusion that the fundamental characters of Cu d-spins at high temperature are unchanged by doping from those of the 2D-Heisenberg system even for the moderately doped high-T_c superconductor $La_{1.85}Sr_{0.15}CuO_4$. This finding eliminates any theoretical models which predict a strong doping dependence of $1/T_1$ around ~ 700 K ($\sim J/2k_B$). At the highest doping level with which we are concerned, 15%, the average distance d between two holes is $d/a \simeq 2.6$. The antiferromagnetic correlation length is less than $\xi/a \simeq 3$ above 700 K for the undoped material as shown in Fig. 5, and neutron scattering experiments showed that hole doping further reduces ξ [4]. A simple physical picture is that as long as the chance of the presence of a doped hole within the radius $\sim\xi$ is not large, the spin dynamics of d-spins does not change dramatically. However, ξ increases with decreasing temperature even in the presence of doped holes as observed by neutron scatting experiments [4,23], and once ξ becomes comparable to d, the doped holes strongly suppress the low-frequency part of the spin fluctuations compared with the undoped case. For $x = 0$, $1/T_1$ at high temperatures is primarily dominated by the reorientation effects of d-spins due to the nearest-neighbor antiferromagnetic exchange interaction J. In view of the fact that $1/T_1$ does not change greatly at high temperatures, strong renormalization of J by the presence of holes is quite unlikely. In other words, spin dynamics in a very short distance scale does not change greatly with doping at high temperatures. However, this does not necessarily mean that the spin stiffness constant ρ_s, a parameter which characterizes how a spin system reacts to a disturbance in a longer scale, is also unchanged by doping.

Below 700 K where the undoped 2D-Heisenberg system shows a divergence of $1/T_1$ due to critical slowing down, hole doping strongly suppresses $1/T_1$. Since the observed temperature dependence of $1/T_1$ is remarkably similar for all of the three insulating samples, one may suspect that the theoretical expressions derived in the renormalised classical regime are valid even for the lightly doped materials. In Fig. 3, the temperature dependence of $1/T_1T^{1.5}$ is compared for $x = 0$, 0.02 and 0.04. Utilizing Eq. (3) and (6a), we might conclude that the systematic decrease of the slope in Fig. 3 is an indication that ρ_s decreases with doping, but we can hardly define a linear region for $x = 0.02$ and 0.04. This finding that renormalization of ρ_s alone is not enough to fit the data is probably related to the observation by neutron scattering experiments by Keimer et al. that the correlation length ξ does not follow Eq. (3) and saturates below 300 K for doped materials [23]. Even if we plug the temperature dependence of ξ observed by Keimer et al. [23] into the expression of $1/T_1$ [18] while choosing ρ_s as a free parameter, the qualitative feature of our data can not be reproduced at low temperatures.

For $x = 0.075$ which superconducts below 23 K, $1/T_1$ slightly increases down to 300 K, and then starts decreasing. This means that the doped system still tries to go into the renormalized classical regime and intends to form an ordered Néel state in the ground state at least down to 300 K. It is quite interesting to notice that the correlation length $\xi/a \simeq 3$ determined by neutron scattering experiments and the average distance between holes $d/a \simeq 3.7$ are comparable at 300 K for this doping level as reported by Birgeneau et al. [4]. Finally at $x = 0.15$, $1/T_1 = 2900 \pm 100\,s^{-1}$ is completely temperature independent down to ~ 350 K, then $1/T_1$ decreases monotonically below 350 K. The temperature independence of $1/T_1$ means that q-averaged $\chi''(q, \omega_n)$ satisfies a Curie law $\langle \chi''(q, \omega_n) \rangle \sim 1/T$

and acts completely paramagnetically. Again, neutron data indicate that $\xi/a \sim d/a$ is satisfied at 300 K [4]. These findings for $x = 0.075$ and 0.15 indicate that the growth of antiferromagnetic correlation tends to saturate with lowering temperature in metallic systems, once the correlation length becomes comparable to the average distance between holes.

Reduction of $1/T_1$ below 350 K for $x = 0.15$ is followed by a drastic power-law like reduction just below T_c [24]. The disappearance of the relaxation mechanism unambiguously indicates that the d-spin degree of freedom is directly involved in the Cooper pairing. Nevertheless d-spins seem to be localized at high temperatures. We therefore suggest that d-spins undergo a gradual transition in nature below ~350 K from localized to itinerant regime. One should also notice that the observed behavior for $x = 0.15$ is qualitatively consistent with the case of the crossover from quantum critical regime to quantum disordered regime [25,12].

6. Conclusions

Firstly, we demonstrated that the magnetic properties of La_2CuO_4 probed by ^{63}Cu NQR can be understood very well based on the 2D-Heisenberg model. Given this result, our observation that $1/T_1$ at high temperatures is virtually the same regardless of the amount of hole doping indicates that the appropriate starting point of the theoretical description of the doped CuO_2 plane in high-T_c cuprates is the 2D-Heisenberg model. In this regard, it is encouraging that recent calculations based on the t–J model reproduce the observed temperature dependence of ξ for doped materials [23] in the quantum critical regime [22]. Our results also imply that the spin dynamics on a short-distance scale does not change significantly by doping at elevated temperatures. When the correlation length becomes comparable to the distance between holes, doped holes significantly limit the growth of spin fluctuations. Since the renormalization of the spin stiffness constant ρ_s alone is not satisfactory to fit the data of $1/T_1$ for doped materials, it seems essentially important to take into account the presence of mobile holes in the theoretical models for doped materials.

Acknowledgements

The authors would like to thank L.P. Gor'kov, A. Sokol, M. Gelfand, S. Chakravarty, R. Birgeneau, M. Kastner, D. Pines, A. Chubukov, P.A. Lee, D. Thelen, M. Klein, M. Salamon, T. Moriya, J.P. Lu, D.E. MacLaughlin and F. Borsa for their helpful discussions and communications. This work was supported by the U.S. National Science Foundation through the Science and Technology Center for Superconductivity Grant No. DMR-91-20000 (C.P.S. and T.I.) and under Grant No. DEFG02-91ER45439 to the University of Illinois at Urbana-Champaign, Frederick Seitz Materials Research Laboratory (C.P.S. and T.I.). The work in Japan was supported by a Grant-in-Aid for Scientific Research on Priority Area 'Chemistry of New Superconductors' (01645004 and 02227102) from the Ministry of Education, Science and Culture of Japan.

References

[1] For general reviews, see Physical Properties of High-T_c Superconductors, Vols. 1–3, ed. D.M. Ginsberg (World Scientific, Singapore, 1991).
[2] P.W. Anderson, Science 235 (1987) 1196.
[3] T. Imai et al., J. Phys. Soc. Jpn. 57 (1988) 1771, 2280.
[4] R. Birgeneau et al., Phys. Rev. B 38 (1988) 6614.
[5] For a review, see C.H. Pennington and C.P. Slichter, in: Vol. 2 of Ref. [1]. Original works on $La_{2-x}Sr_xCuO_4$ include Refs. [6,8,9,24].
[6] T. Imai et al., J. Phys. Soc. Jpn. 59 (1990) 3846.
[7] M. Mali et al., Physica C 175 (1991) 58.
[8] T. Imai, C.P. Slichter, K. Yoshimura and K. Kosuge, Phys. Rev. Lett. 70 (1993) 1002.
[9] T. Imai, C.P. Slichter, K. Yoshimura, M. Katoh and K. Kosuge, Phys. Rev. Lett. 71 (1993) 1254.
[10] S. Chakravarty, B.I. Halperin and D.R. Nelson, Phys. Rev. B 39 (1989) 2344; S. Tyč, B.I. Halperin and S. Chakravarty, Phys. Rev. Lett 62 (1989) 835.
[11] A.V. Chubukov and S. Sachdev, Phys. Rev. Lett. 71 (1993) 169; S. Sachdev and J. Ye, Phys. Rev. Lett. 69 (1992) 2411.

[12] A. Sokol and D. Pines, Phys. Rev. Lett. 71 (1993) 2813.
[13] T. Moriya, J. Phys. Soc. Jpn. 18 (1963) 516.
[14] F. Mila and T.M. Rice, Physica C 157 (1989) 561.
[15] C.H. Pennington et al., Phys. Rev. B 39 (1989) 274.
[16] C.H. Pennington and C.P. Slichter, Phys. Rev. Lett. 66 (1991) 381. Also, see D. Thelen and D. Pines, preprint.
[17] P. Hasenfatz and F. Niedermayer, Phys. Lett. B 268 (1991) 231.
[18] S. Chakravarty and R. Orbach, Phys. Rev. Lett. 64 (1990) 224.
[19] S.M. Hayden et al., Phys. Rev. Lett. 67 (1991) 3022.
[20] M.P. Gelfand and R.R.P. Singh, Phys. Rev. B 47 (1993) 14413.
[21] A. Sokol, E. Gagliano and S. Bacci, Phys. Rev. B 47 (1993) 14646.
[22] A. Sokol, R. Glenister and R.P.P. Singh, preprint.
[23] B. Keimer et al., Phys. Rev. B 46 (1992) 14034.
[24] K. Ishida et al., J. Phys. Soc. Jpn. 58 (1989) 2508; K. Yoshimura et al., J. Phys. Soc. Jpn. 59 (1989) 3073; Y. Kitaoka et al., Physica C 170 (1990) 189.
[25] S. Chakravarty, private communication.